Rethinking of the Evolution Process in Genetic Programming

Schema Theory and its new behavioral (Semantic) version in Genetic Programming

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Evolutionary Algorithms

Genetic Algorithm (GA), Genetic Programming (GP), Gene Expression Programming (GEP)

- Schema: Co-dependency Relationship
- Genetic Programming
- Challenges of modeling Schema in GP
- Valued Schema
- Valued Schema Theory
- Experiment Results

Evolutionary Algorithms as Perturbation-Based Schema Testing

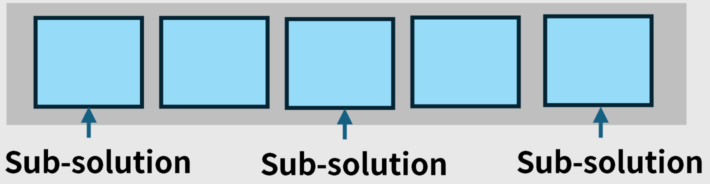
Flow of Evolutionary Algorithms

(Genetic Algorithm, Genetic Programming, Gene Expression Programming)

Initialize Population → Evaluation → Selection and Reproduction → Genetic Variation → Evaluation

Individual – Candidate Solution

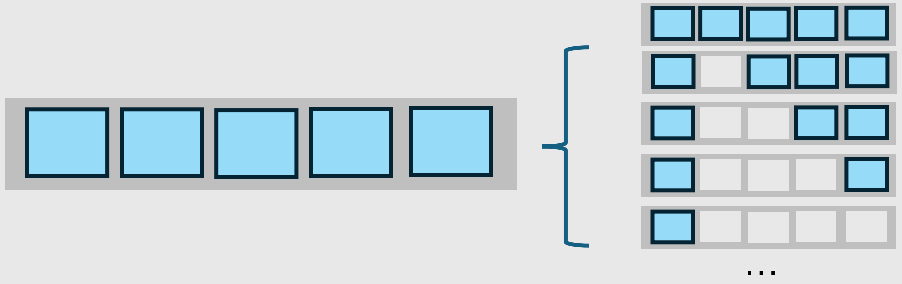
Sub-structures →



Schema: Co-dependency relationship of sub-solutions

Which sub-solutions need to co-occur in specific structural positions within an individual?

One individual contains millions of different levels schema.



Evolutionary Algorithms implicitly and massively test Schemas by testing individuals in parallel.

Goal of Evolutionary Algorithms

Find Stable and Contributing Schema.

Schema Theory

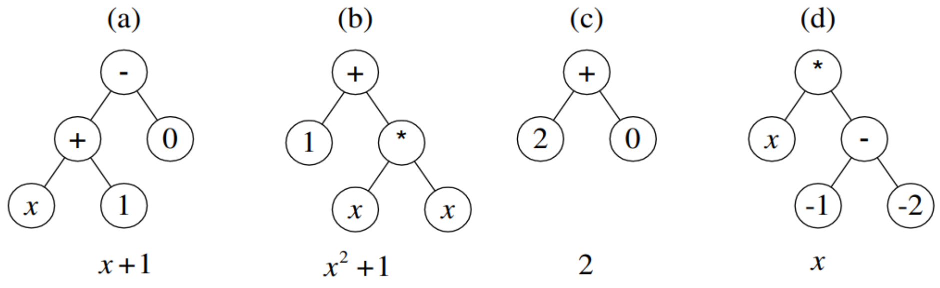
A mathematical model for dynamics of schema during evolution.

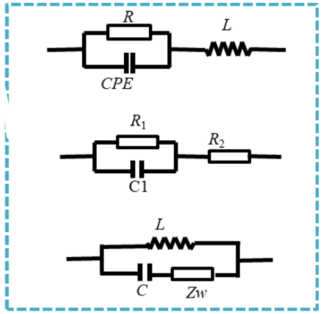
$$P(H,t+1) \geq P(H,t)\underbrace{[amplify(H,t)]}_{\text{Effect from Selection}}\underbrace{[1-disruption(H)]}_{\text{Effect from Genetic Variation}}$$

Genetic Programming

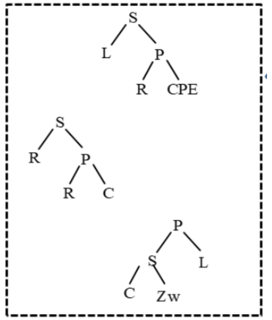
Individuals Presentation

One Solution = One Tree





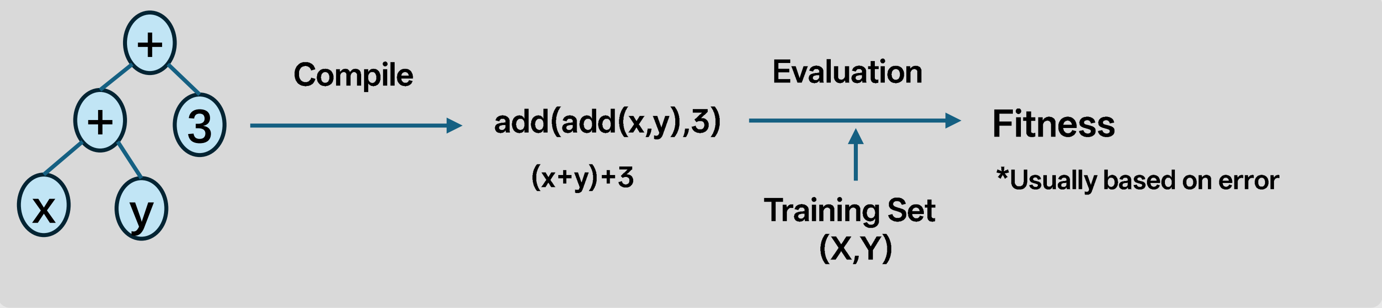
d) Electrical Equivalent Circuit



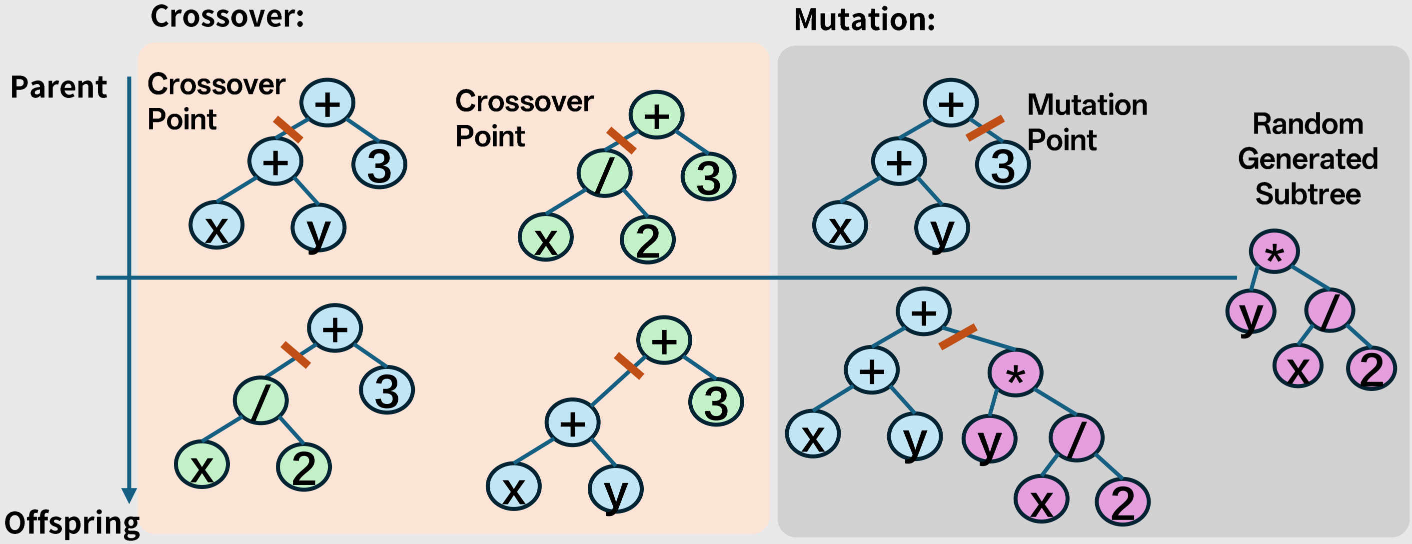
c) Binary tree

```
…(If-Bumped:Turn-
Right(Case-
Radar:Turn-Right-
Turn-Right-
Spd:Spd))(Case-
Touch:Grasp:Turn-
Right:Release:Turn-
Left:Turn-Right)…
```

Selection is purely Behavioral-based.



Genetic Variation is purely Structural-based.



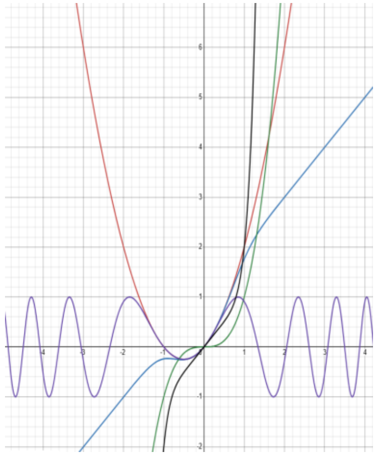
This dual-space mechanism makes GP powerful but difficult to analyze.

Challenges of Constructing Schema Theory in GP

Non-aligned Mapping between Structure and Behavior

Similar Structure → Different Behavior

```
add(power(x,2),x)
add(tanh(x,2),x)
mul(power(x,2),x)
sin(power(x,2),x)
add(power(x,8),x)
```



Img: https://www.desmos.com/

Similar Behavior → Different Structure

$\sin(x)$

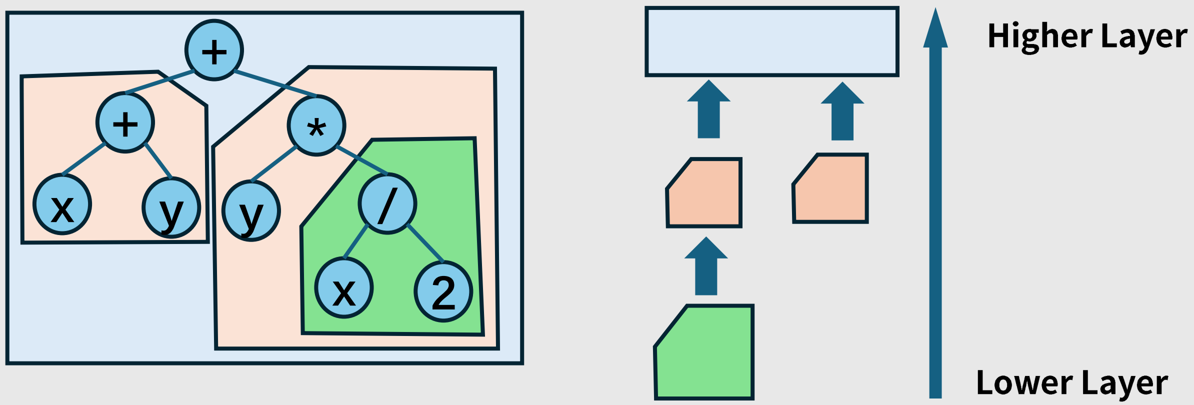
$\sin(x) + (x - x) \times (x^2 + x + 1)$

$\sin(x) + [(x^2 + 2x) - (x^2 + 2x)] \times (\sqrt[2]{\frac{x+1}{x}})$

Observation in Behavior Space

- Selection is evaluated on semantic outputs.
- Selection pressure propagates behavioral signals.

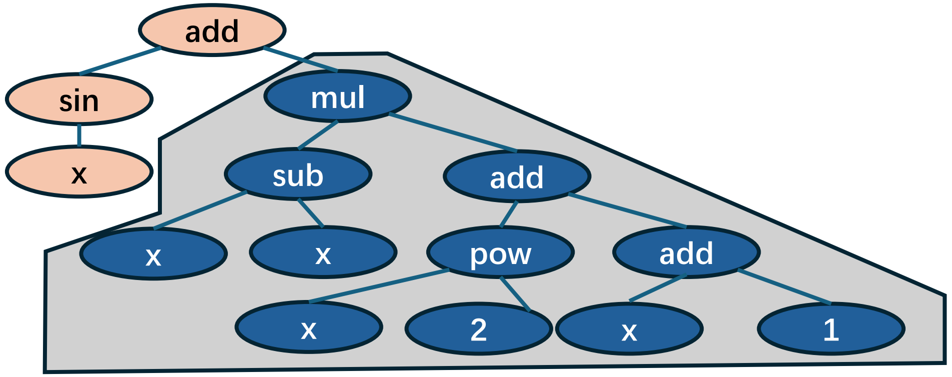
Computation Graph:



Abstraction of Schema:
Difficult in Structural but possible in Behavioral

Schema in Behavioral Space

Semantic Decomposition of a tree (Subtree)



$\sin(x) + (x - x) \times (x^2 + x + 1)$

Dominant Part:

- Semantic valid part
- Contribute to behavioral output of a tree

Zero Part:

- Semantically neutral region
- No contribution to the output under current semantics

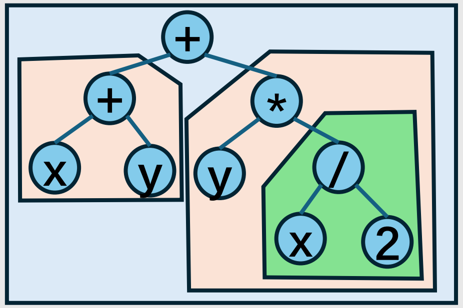
Behavioral Decomposition:

$$\mathrm{V(Tree)} = \mathrm{V(Dominant)} + \mathrm{V(Zero)} = \mathrm{V(Dominant)}$$

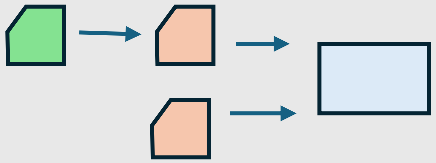
Valued Schema

A set of subtrees that satisfy:

- Identical semantic behavior
- Identical dominant size (number of dominant symbols)



Computation Graph:



GP implicitly tests valued schemas through parallel evaluation of individuals.

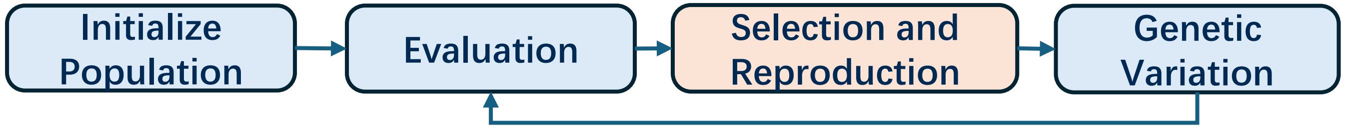
Dynamics of Valued Schema

Schema Theory

Conceptual lower bound of schema propagation:

$$P(H,t+1) \geq P(H,t)\underbrace{[amplification(H,t)]}_{\text{Effect from Selection}}\underbrace{[1-disruption(H)]}_{\text{Effect from Variation}}$$

$P(H,t)$: The ratio of schema H in current population



Effects from Selection

Long-term optimal rule of Exploration and Exploitation to design rational selection mechanism:

Probability of Schema/individual being selected is proportional to its relative fitness.

$$amplification(H,t) = \frac{f(H)}{\bar{f}(t)}$$

Fitness of all individuals carries schema H

Average fitness of current population

Result:

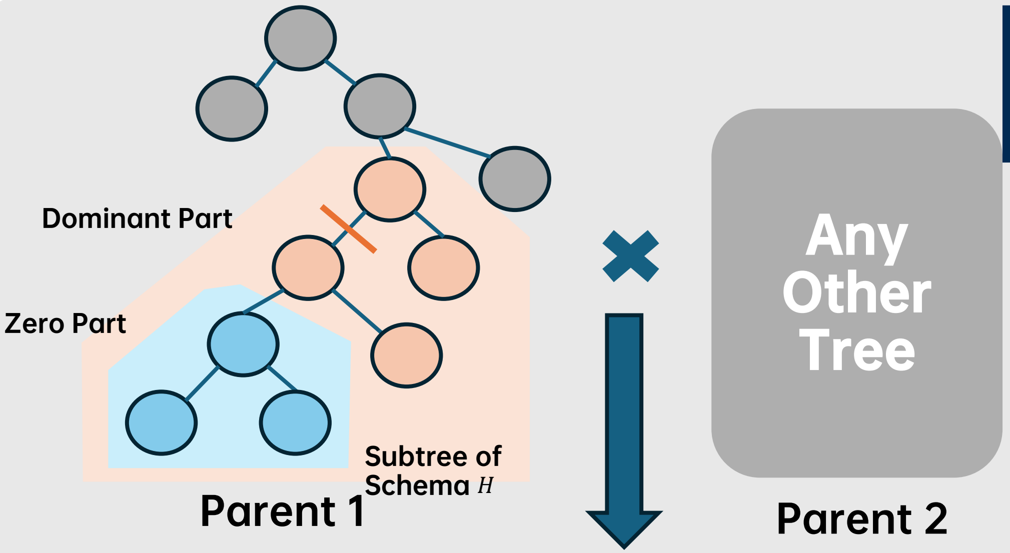
Schemas above average are amplified; below average are suppressed proportionally.

Dynamics of Valued Schema (cont'd)

Effects from Crossover

$$P(H,t+1) \geq \underbrace{P(H,t)\frac{f(H)}{\bar{f}(t)}}_{\text{Effect from Selection}}\underbrace{[1 - disruption(H)]}_{\text{Effect from Variation}}$$

Initialize Population → Evaluation → Selection and Reproduction → Genetic Variation → Evaluation



Schema survives in crossover disruption or not?

Functionality of Crossover

Block disruption-based test of schema

Relationship Disruption
- Important to solve problem → Fitness drop → Extinction
- Unimportant → Fitness almost unchanged → Survive

Result:
Only Important Schema will be left out.

Consider All possibilities:

Crossover point locates on the subtree of Schema
- The other parents?
  - From the same Schema
  - From the different Schema
- Crossover Point Location
  - On Dominant part
  - On Zero part

$$P_{disrupt}(H,t) = E_{size(H,t)}\left[\text{Dominant term}\big(k(H,t), r(H)\big) + \text{Zero term}\big(\eta(H,t), 1 - r(H)\big)\right]$$

Complete Version of Valued Schema Theory

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$$P(H,t+1) \geq P(H,t)\underbrace{[amplification(H,t)]}_{\text{Effect from Selection}}\underbrace{[1 - disruption(H)]}_{\text{Effect from Variation}}$$

Discussion about crossover
(Be crossovered or not)

$$P(H,t+1) \geq P(H,t)\frac{f(H,t)}{\bar{f}}\Bigg[\underbrace{(1-p_c)}_{none-xover} + \underbrace{p_c(1-P_{disruption})}_{xover}\Bigg]$$

$$P(H,t+1) \geq P_{selection}(H,t)[(1 - p_c P_{disruption})]$$

$P(H,t)$: The ratio of schema H in current population

p_c: Crossover Rate (Hyper-param)

Effect from Selection

$$P_{selection}(H,t) = P(H,t)\frac{f(H)}{\bar{f}}$$

Effect from Variation (Crossover Only, Full version)

$$P_{discrupt}(H,t) = E_{\text{size}}(H,t)\Bigg[\underbrace{(1 - P(H,t)k(H,t))r(H)}_{\text{Dominant}} + \underbrace{\Big((1-P(H,t))\eta_1 + P(H,t)\eta_2\Big)(1-r(H))}_{\text{Zero}}\Bigg]$$

Full Expansion of Valued Schema Theory

$$P(H,t+1) \geq P(H,t)\frac{f(H,t)}{\bar{f}}\Bigg[1 - p_c E_{size}(H,t)\Bigg(\underbrace{\left(1 - P(H,t)\frac{f(H,t)}{\bar{f}}k(H,t)\right)r(H)}_{\text{Dominant}} + \underbrace{\left(\left(1 - P(H,t)\frac{f(H,t)}{\bar{f}}\right)\eta_1(H,t) + P(H,t)\frac{f(H,t)}{\bar{f}}\eta_2(H,t)\right)(1-r(H))}_{\text{Zero}}\Bigg)\Bigg]$$

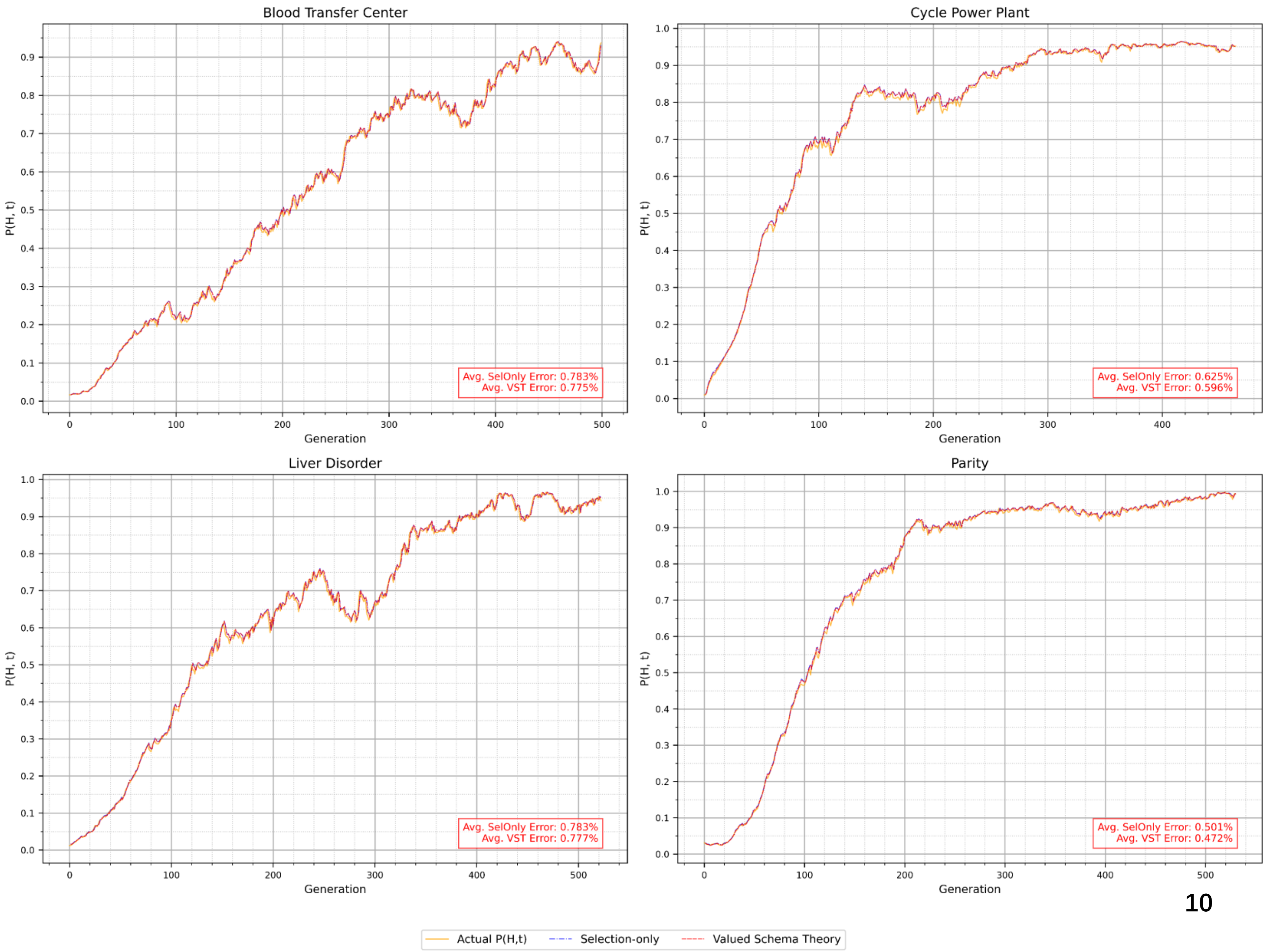
Experiment Result of Prediction GP's Behavior

Use Valued Schema Theory to **predict real GP behavior** during solving problems.

Prediction of GP's behavior

- Track most schemas changing dynamics.
- Test on 4 different problems in different categories [Classification, Regression, Continuous, Discrete].

All Errors between Valued Schema Theory and real GP dynamics: **<1%**



What we do?

- Developed a new Schema Theory of GP in a behavior perspective: Practically Trackable, Highly Accurate

What can it do?

- Offer a new perspective on how GP find optimal solution, and Provide insights to other GP algorithm developments :

e.g., Value-aware subtree Transfer Learning GP
Parallel Computation
Anti-bloat Techniques

What can you bring to home?

- Evolutionary Algorithms are explainable by Schema Theory.

- GP(or other EAs) test individuals, and meanwhile massively tests schemata (co-dependency relationship of sub-solutions) to solve the problem.

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Reference

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2. R. Poli, W. B. Langdon, and N. F. McPhee, *A Field Guide to Genetic Programming* (Lulu Enterprises, UK Ltd, 2008).
3. U.-M. O'Reilly and F. Oppacher, The troubling aspects of a building block hypothesis for genetic programming, Found Genet Algorithms **3**, 73 (1995).

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Takuya Ito, Hitoshi Iba, and Masayuki Kimura. 1996. Robustness of robot programs generated by genetic programming. In Proceedings of the 1st annual conference on genetic programming. MIT Press, Cambridge, MA, USA, 321–326.

Related Publications

1. Yilin Liu, Zhengwen Huang. Semantic Valued Schema Theory of Genetic Programming in Symbolic Regression, 23 December 2025, PREPRINT, available at Research Square [https://doi.org/10.21203/rs.3.rs-8316711/v1]
2. Y. Liu, G. Taylor, and Z. Huang, Novel application of mutual information in transfer learning for genetic programming. In: Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO Companion) pp. 635–638. Association for Computing Machinery, Malaga, Spain (2025).